CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EE)

November 2012

EE 305 Power Electronics & Drives - I

Date: 30.11.2012, Friday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Explain V-I Characteristics of G.T.O. Also explain disadvantages of G.T.O. [05]

(b) State different types of Power Electronic Converters. [02]

Q - 2 (a) Explain briefly Switching Characteristics of Thyristor. [04]

(b) Explain with a neat diagram Static V-I Characteristics of a Thyristor. [04]

(c) State and Explain briefly different types of ratings for Thyristor. [06]

OR

Q - 2 (a) Write a Short note on protection of Thyristor. [04]

(b) What is String efficiency? Explain briefly parallel operation of SCRs. [06]

(c) It is required to operate 250 A SCR in parallel with 350 A SCR with their respective on-state voltage drops of 1.6 V and 1.2 V. Calculate the value of resistance to be inserted in series with each SCR so that they share the total load of 600 A in proportion to their current ratings. [04]

Q - 3 (a) Explain with neat waveforms operation of UJT based Ramp triggering circuit. Also explain the effect of variation in charging resistor on output pulses. [06]

(b) Explain in detail driver circuit for BJT [04]

(c) Explain in detail Complementary Commutation technique for Thyristor. [04]

OR

Q - 3 (a) What is the difference between triggering and commutation of Thyristor? Explain briefly External Pulse Commutation technique used for Thyristor. [04]

(b) Explain with neat waveforms resistance firing of an SCR. [04]

(c) List out the requirements of a BJT driver circuit. Explain driver circuit for MOSFET. [06]

SECTION – II

Q - 4 (a) Explain with output voltage and current waveforms operation of step down chopper. [04]

(b) A step-up chopper has input voltage of 220 V and output voltage of 660 V. If the non-conducting time of thyristor-chopper is 100 μs, compute the pulse width of output [03] voltage. In case pulse width is halved for constant frequency operation, find the new output voltage.

Q - 5 (a) Explain operation of Two-quadrant type-A and Two-quadrant type-B Chopper [07]

(b) Explain in detail with neat waveforms operation of 1-Φ, Full wave, Full converter controlled DC series motor. [07]

OR

Q - 5 (a) Explain in detail with neat waveforms chopper control of separately excited DC Motor. [07]

(b) Explain with neat waveforms operation of Current-Commutated Chopper. [07]

Q - 6 (a) Explain in detail with neat waveforms of Supply Voltage, Load Voltage, Load Current and Thyristor Current, operation of 3-Φ, Full Wave, Fully Controlled Bridge Rectifier with R-L load for firing angle $\alpha = 30^0$ & 90^0. [07]

(b) A 1-Φ, 230 V, 1 KW heater is connected across 1-Φ, 230 V, 50 Hz supply through an SCR. For firing angle delays of 45^0 and 90^0, calculate the power absorbed in the heater element. [03]

(c) Write Short note on Extinction angle control scheme for power factor improvement of Phase controlled rectifiers. [04]

OR

Q - 6 (a) Explain with neat waveforms the operation of 1-Φ, Semiconverter with R-L load for continuous load current. Also derive equation of Average and RMS value of output voltage. [07]

(b) Explain in detail operation of Dual Converter with and without circulating current. [07]

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